ASSESSMENT - 1

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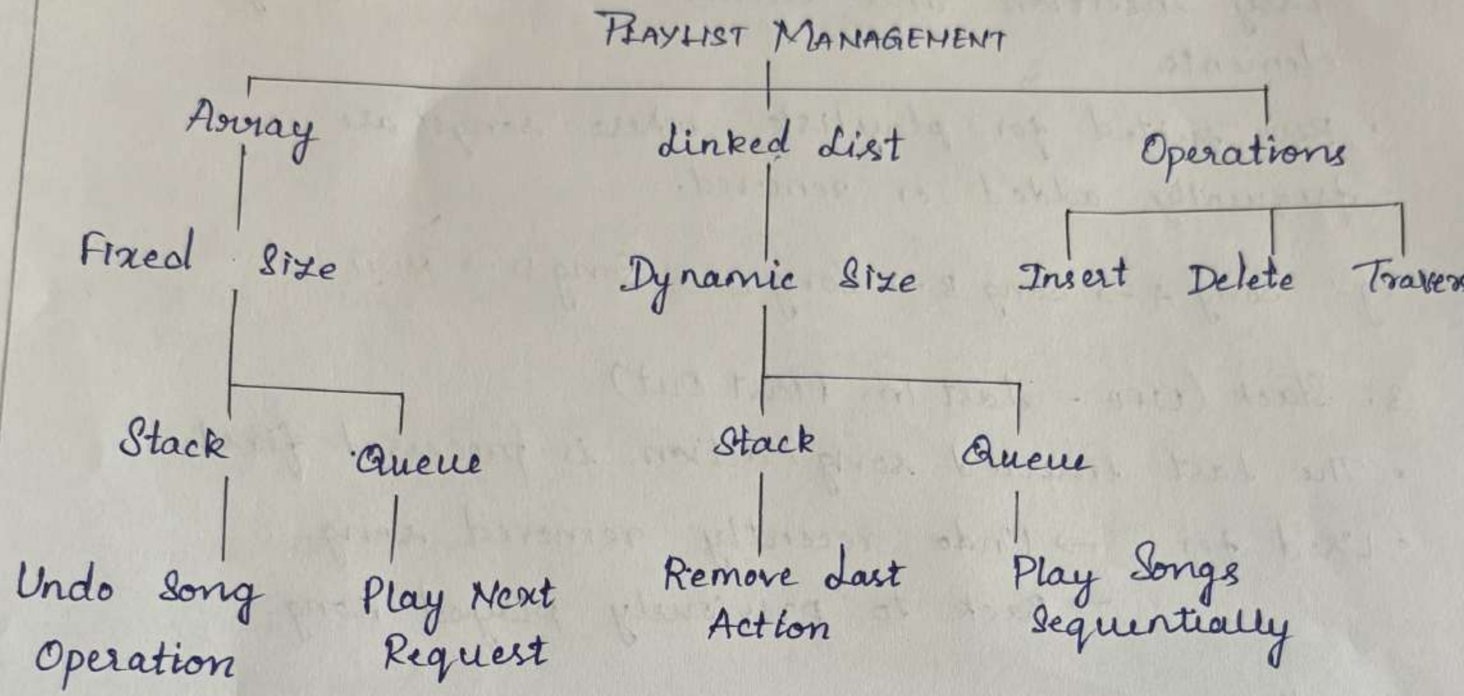
9)

Concept Mapping

Playlist Management Using Data Structures.



Explanation

1. Array

- Stores songs in contiguous memory.
- Supports fast access using index.
- Insertion and deletion are slow because elements must be shifted.
- Suitable for small playlists with a fixed number of songs.

eg: [Song A] [Song B] [Song C] [Song D]

2. Linked List

- Stores songs as nodes connected by pointers.
- Dynamic in size.
- Easy insertion and deletion without shifting elements.
- Best suited for playlists where songs are frequently added or removed.

eg: Song A → Song B → Song C → Song D → NULL.

3. Stack (LIFO - Last In, First Out)

- The last inserted song / action is processed first.
- Used for: → Undo recently removed song.
  → Back to previously played song.

Operations

- Push - Add action
- Pop - Undo last action

4. Operations		Purpose in Playlist
Insert	→	Add a new song
Delete	→	Remove a song
Traverse	→	Display or play all songs sequentially.

14) Ticket Booking System

TICKET BOOKING SYSTEM

Abstract Data Type

Operations

Implementation

Booking

Cancellation

Seat Allo-cation

Array

Linked List

Data Structures

Array

- Fixed Size
- Storage Direct
- Index Access
- Fast Search

Linked List

- Dynamic Storage
- Node-based Storage
- Easy Insert / Delete

Performance Analysis

Array Implementation

- Booking → O(1)
- Cancellation → O(1)
- Seat Search → O(1)
- Insert / Delete → O(n)

Linked List Implementation

- Booking → O(n)
- Cancellation → O(n)
- Seat Search → O(n)
- Insert / Delete → O(1)

Best choice for Real-Time Booking

Array Implementation

- Fast seat looking using index
- Quick booking and cancellation
- Better cache performance
- Suitable for fixed number of seats
- Used in theatres, buses and flights